

JWALINSHAH

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AI Systems Engineer — Agent Reliability · Evaluation · Local Inference

Builds systems that constrain, observe, and verify probabilistic agents under real-world constraints.

PROFESSIONAL EXPERIENCE

tinyhumansai | Core Open-Source Contributor — OpenHuman

Mar 2026 – Present

- Landed 69 merged upstream PRs across a Rust/TypeScript desktop agent, spanning memory citations and namespaces, typed controller/RPC registries, privacy and security surfaces, chat reliability, and regression coverage.
- Shipped SQLite-backed contact identity resolution and scoring with CNContactStore; moved ~1,500 lines of CLI adapters into domain-owned Rust modules and added 196 tests across MCP, services, utilities, and autocomplete.
- Designed and prototyped an on-device personal index using FTS5 + sqlite-vec hybrid retrieval with staged iMessage, Gmail, Calendar, Contacts, and chronicle ingestion.

Break the Web | AI Systems Engineer — LiveLM

Remote · May 2026 – Present

- Built current-world retrieval over ~80K indexed canonical facts with explicit ok / partial / stale / miss routing and ~11 ms warm p50, forcing fallback instead of plausible stale answers.
- Built a 71-question × 3-serving-path benchmark plus 512 story-anchored probes and sidecar verification across retrieval, freshness, answerability, and fallback behavior.

Skill AI | Data Operations Lead — ML Systems & Reliability

San Mateo, CA · Jul–Nov 2025

- Built collection, validation, and evaluation systems across five robotic platforms; Python and Apps Script automation cut administrative overhead ~50% and improved downstream task success ~40%.
- Scaled a 30+ operator program with scheduling, QA, training, compliance, and runbooks; led technical execution for 25+ live demos during a \$1B Series C with zero demo failures.

SELECTED SYSTEMS

Bridge — Reliable Coding-Agent Execution

Go · Seatbelt · Worktrees · Formal Methods

- Built an isolated spawn → verify → deliver runtime with allowlisted adapters, deny-default macOS sandboxing, leased worktrees, and fresh-checkout verification so workers cannot certify their own outputs.
- Cataloged 77 invariants and checked or proved 52 using 28 property-based checkers, 8 Z3 proofs, 5 TLA+ models (including a 1.58M-state exploration), and Lean proofs.

OfficeQA Arena — Grounded Reasoning Evaluation

Python · Retrieval · Tool Use

- Reached 184.5/246 (75%) on a 246-task Sentient benchmark for \$1.71 after 12 architecture rounds and ~4,400 evaluations.
- Trace analysis showed evidence selection caused 48% of failures, while correctly grounded Python answers had 0% arithmetic errors — retrieval and tool selection, not model arithmetic, were the dominant bottlenecks.

Voice Engine — On-Device Speech Infrastructure

Swift · CoreML · MLX

- Built a fully local dictation runtime with Moonshine ASR, VAD, FullStop punctuation, deterministic correction, and native text injection; 348/349 tests pass, with one isolated macOS CoreML state regression.
- Built WER and real-time-factor harnesses to compare model/configuration tradeoffs on Apple Silicon and isolate transcription, punctuation, vocabulary, and runtime failures.

TECHNICAL SKILLS

Languages: Go, Rust, Python, Swift, TypeScript, SQL

Systems & AI: Agent reliability and evals; BM25/vector/hybrid retrieval; tool-augmented LLMs; local inference; sandboxing;

formal verification and model checking; benchmark, trace, and failure-mode analysis

Technology: MLX, CoreML, FastAPI, SQLite/sqlite-vec, TLA+, Lean, Z3, Docker, GCP, Git

EDUCATION & LEADERSHIP

University of Texas at Dallas — B.S. Computer Science, Dec 2024

Ultimate Frisbee Captain — led a 30-member program · UTD Chess Club — member